

Arunabha Majumder

Mechanical R&D Engineer | Mechanisms & Wearable Robotics | Prototyping | Test Rigs & Validation

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PROFESSIONAL SUMMARY

Mechanical R&D Engineer with a B.Tech in Mechanical Engineering, an M.Tech in Mechatronics and a PhD in progress in Denmark, developing wearable load-carrying mechanisms that people use on their bodies. Takes mechanical concepts from whiteboard idea to tested hardware: mechanism development, SolidWorks CAD, FEA and structural analysis, rapid prototyping from machined and 3D-printed parts, custom test rigs, functional testing and validation with human users. Curious about how mechanisms work and why they fail, with hands-on engineering projects ranging from a self-built Go-Kart to hand-made pneumatic muscles. Best Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA).

CORE SKILLS

- **Mechanisms & Concepts:** Mechanism development, moving assemblies, compliant and variable-stiffness actuation, new mechanical concepts, load-bearing wearable structures
- **Mechanical Design & Analysis:** SolidWorks (CSWA certified), Fusion 360, Finite Element Analysis (FEA), analytical stiffness and kinematic modelling, structural behaviour
- **Prototyping & Manufacturing:** Rapid prototyping, proof-of-concept builds, machined parts, additive manufacturing / 3D printing, hands-on fabrication and assembly, Design for Assembly (DFA)
- **Testing & Verification:** Test fixture and test rig design, functional testing, verification and validation, user testing, experimental data analysis
- **Problem Solving:** Analytical problem solving, model-to-hardware validation, design iteration from test results
- **Mechatronics & Tools:** Motor, encoder, force-sensor and DAQ integration, pneumatic actuators, embedded control, Python, MATLAB/Simulink

PROFESSIONAL EXPERIENCE

Research Engineer – Mechanisms & Wearable Robotics R&D (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Developed a hybrid-actuated shoulder exoskeleton from concept to tested hardware: mechanism concept, SolidWorks CAD, FEA iterations for stiffness, strength and weight, fabrication, assembly and testing.
- Developed a hybrid actuation principle that pairs a parallel spring with a 6 Nm motor to carry quasi-static gravity load; cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Built rapid prototypes hands-on from machined and additively manufactured parts, and integrated motor, encoder, force sensors and DAQ.
- Designed custom automated test rigs and validated analytical stiffness models of variable-stiffness actuators against experimental hardware to below 10% RMS error, de-risking mechanisms before fabrication.
- Removed the external force/torque sensor from a 5-bar parallel robot by deriving force from the actuator's torque–deflection characteristic, reducing cost, mass and integration complexity.
- Validated the device with users in a 12-participant study across 3 assistance modes and 2 load conditions (EMG, motion capture), reducing muscle effort by more than 15%.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

M.Tech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated pneumatic artificial muscle actuators in-house from raw materials and built the test setup with valves, pressure sensors and DAQ to characterise load–displacement behaviour.
- Derived an empirical actuator model from test data and designed a neural-network-based controller, reducing position-tracking error to 0.3–0.78% under varying load.

Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019
Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, fabrication and user testing of a rehabilitation exoskeleton end to end; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

HANDS-ON ENGINEERING PROJECTS

- **Go-Kart (2017):** Designed, fabricated and built a custom Go-Kart from the ground up.
- **Efficycle (2018):** Worked in a multidisciplinary team to design and build a human-electric hybrid vehicle for the Efficycle competition.
- **Sketching:** Diploma in Fine Arts (Painting); freehand sketching for concepts and technical illustration.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022
AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019
Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

CERTIFICATIONS & AWARDS

- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Major Project Award (2019); TATA Technologies Innovation Award – INR 100,000 Grant (2019)

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer.
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

TECHNICAL TOOLS

- **CAD & Manufacturing:** SolidWorks, Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Software & Modelling:** Python, MATLAB/Simulink, LabVIEW (basic), Arduino
- **Testing & Instrumentation:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Pneumatic Control, Custom Rig Fabrication

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.